Addition of angular momenta

$$\vec{J}_1 + \vec{J}_2 = \vec{J}$$

Naturally, the sum of angular momenta is also an angular momentum.

What if we consider $\vec{T} = a\,\vec{J}_1 + b\,\vec{J}_2$?

By definition, angular momentum $\vec{J}$ satisfies

$$[J_i, J_j] = i\hbar\, \varepsilon_{ijk} J_k$$

Let us check $\vec{T}$ now:

$$[T_i, T_j] = [a J_{1i} + b J_{2i}, a J_{1j} + b J_{2j}] =$$

$$= a^2 [J_{1i}, J_{1j}] + b^2 [J_{2i}, J_{2j}] = \left(\text{because } [\vec{J}_1, \vec{J}_2] = 0\right)$$

$$= a^2 i \varepsilon_{ijk} J_{1k} + b^2 i \varepsilon_{ijk} J_{2k} =$$

$$= i \varepsilon_{ijk} \left(a^2 J_{1k} + b^2 J_{2k}\right) \neq i \varepsilon_{ijk} \underbrace{\left(a J_{1k} + b J_{2k}\right)}_{T_k}$$

The only possibility : $a = b = 1$.

$$\left(\text{or } a = 1, b = 0 \text{ or } a = 0, b = 1\right)$$

<u>Eigenfunctions</u>

$$\left.\begin{aligned} \hat{J}_1^2 |j_1, m_1\rangle &= \hbar^2 j_1(j_1+1) |j_1, m_1\rangle \\ \hat{J}_1^z |j_1, m_1\rangle &= \hbar m_1 |j_1, m_1\rangle \end{aligned}\right\} \begin{aligned} &(2j_1+1) \text{ functions} \\ &-j_1 \le m_1 \le j_1 \end{aligned}$$

$$\left.\begin{aligned} \hat{J}_2^2 |j_2, m_2\rangle &= \hbar^2 j_2(j_2+1) |j_2, m_2\rangle \\ \hat{J}_2^z |j_2, m_2\rangle &= \hbar m_2 |j_2, m_2\rangle \end{aligned}\right\} \begin{aligned} &(2j_2+1) \text{ functions} \\ &-j_2 \le m_2 \le j_2 \end{aligned}$$

$\Rightarrow$ $\vec{J} = \vec{J}_1 + \vec{J}_2$ has a $D = (2j_1+1)\cdot(2j_2+1)$ dimensional space of functions $|j_1, m_1; j_2, m_2\rangle = |j_1, m_1\rangle |j_2, m_2\rangle$

Clearly $J^z = J_1^z + J_2^z$: $\hat{J}^z |j_1, m_1; j_2, m_2\rangle = \hbar(m_1 + m_2) |j_1, m_1; j_2, m_2\rangle$

thus eigenvalue of $\hat{J}^z$: $\hbar m$, with $m = m_1 + m_2$.

Note that $m_{max} = m_{1max} + m_{2max} = j_1 + j_2$

which defines unique state $|j_1, j_1; j_2, j_2\rangle$.

Since for arbitrary angular momentum $m_{max} = j$,

it follows that $|j_1, j_1; j_2, j_2\rangle$ is a state of angular momentum $j = j_1 + j_2 \Rightarrow$ hence $\hat{J}^2 |j_1, j_1; j_2, j_2\rangle = \hbar^2 (j(j+1)) \cdot |j_1, j_1; j_2, j_2\rangle$.

Then by acting on this state with $\hat{J}^- = \hat{J}_1^- + \hat{J}_2^-$ repeatedly, we generate $[2(j_1+j_2)+1]$ states, each of angular momentum $j = j_1 + j_2$ but with different m's:

$-(j_1+j_2) \le m \le j_1 + j_2$. Of all these states, only <u>one</u> has $m = j_1 + j_2 - 1$ (every $|m| \le j_1 + j_2$ occurs only once).

Such a state obtains as $\hat{J}^- |j_1, j_1; j_2, j_2\rangle =$

$$= (\hat{J}_1^- + \hat{J}_2^-) |j_1, j_1; j_2, j_2\rangle \sim \sqrt{2j_1}\, |j_1, j_1-1; j_2, j_2\rangle + \sqrt{2j_2}\, |j_1, j_1; j_2, j_2-1\rangle = |\varphi_+\rangle.$$

However there is another function,

$$|\varphi_-\rangle = \sqrt{2j_2}\, |j_1, j_1-1; j_2, j_2\rangle - \sqrt{2j_1}\, |j_1, j_1; j_2, j_2-1\rangle$$

Note that $\langle \varphi_+ | \varphi_- \rangle = 0$. Note also that

$$\hat{J}^z |\varphi_-\rangle = (\hat{J}_1^z + \hat{J}_2^z) |\varphi_-\rangle = \hbar (j_1 - 1 + j_2) \sqrt{2j_2}\, |j_1, j_1-1; j_2, j_2\rangle$$

$$- \hbar (j_1 + j_2 - 1) \sqrt{2j_1}\, |j_1, j_1; j_2, j_2-1\rangle = \hbar (j_1 + j_2 - 1) |\varphi_-\rangle.$$

Hence $m_{\varphi_-} = j_1 + j_2 - 1$.

Also, $\hat{J}^+ |\varphi_-\rangle = (\hat{J}_1^+ + \hat{J}_2^+) |\varphi_-\rangle \sim \sqrt{2j_2 \cdot 2j_1}\, |j_1, j_1; j_2, j_2\rangle - \sqrt{2j_1 \cdot 2j_2}\, |j_1, j_1; j_2, j_2\rangle$

$= 0$.

$\Rightarrow \max(m_{\varphi_-}) = j_1 + j_2 - 1 \Rightarrow$ this function belongs to $j = j_1 + j_2 - 1$ manifold.

$J^- |\ell, m\rangle = \hbar \sqrt{(\ell - m + 1)(\ell + m)}\, |\ell, m-1\rangle$

$J^+ |\ell, m\rangle = \hbar \sqrt{(\ell - m)(\ell + m + 1)}\, |\ell, m+1\rangle$

Hence by applying $\hat{J}^-$ to $|\varphi_-\rangle$ repeatedly, we will generate $[2(j_1+j_2-1)+1]$ different values of m.

Look now on $m = j_1+j_2-2$ which can be achieved in 3 ways: $m_1 = j_1-2,\ m_2 = j_2$; $m_1 = j_1-1,\ m_2 = j_2-1$; $m_1 = j_1,\ m_2 = j_2-2$. Two possibilities have already being accounted for via $|\varphi_+\rangle$ and $|\varphi_-\rangle$ states. The <u>third</u>, orthogonal to both $|\varphi_\pm\rangle$, can be used to construct the manifold with $j = j_1+j_2-2$.

This procedure can be continued until all $(2j_1+1)(2j_2+1)$ original states have been used. This happens when $j = |j_1-j_2|$. (Indeed, assume $j_1 > j_2$ for definiteness: than the minimal possible value of $m_{max} = j_1-j_2$. Hence the "minimal" manifold must have $j = j_1-j_2$.)

Thus, upon adding $\vec{J}_1$ (with $\hat{J}_1^2 \to \hbar^2 j_1(j_1+1)$) and $\vec{J}_2$ (with $\hat{J}_2^2 \to \hbar^2 j_2(j_2+1)$) we get ~~states~~ manifolds with angular momenta $\boxed{j_1-j_2 \le j \le j_1+j_2}$. Each has $(2j+1)$ ~~degenerate~~ with different m's.

Total number of them $\sum_{j_1-j_2}^{j_1+j_2}(2j+1) = (2j_1+1)(2j_2+1)$.

Check it: $\sum_{j_1-j_2}^{j_1+j_2}(2j+1) = \underbrace{2(j_1-j_2)+1}_{1} + \underbrace{2(j_1-j_2+1)+1}_{2} + \dots + \underbrace{2(j_1+j_2)+1}_{2j_2+1}$

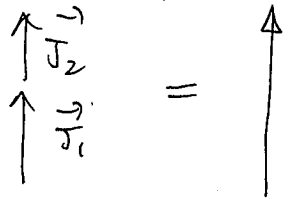
(the sum has $(2j_2+1)$ terms) $=$

$$= (2j_1)\cdot(2j_2+1) + (1)\cdot(2j_2+1) = (2j_1+1)(2j_2+1).$$

[Note that every entry with $-j_2+k$ $(0 \le k \le 2j_2+1)$ is compensated by a symmetric one with j_2-k.)

Very clear classical analogy: when adding vectors $\vec{J}_1$ and $\vec{J}_2$, the maximum length vector has $j = j_1 + j_2$,
(j_1) (j_2) the minimum length $j = j_1 - j_2$.

<u>parallel</u>



$= \uparrow \quad \underline{\vec{J} = \vec{J}_1 + \vec{J}_2}$

<u>antiparallel</u>

$= \uparrow \quad \underline{\vec{J} = \vec{J}_1 - \vec{J}_2}$

Denote the new set of states as
$|j, m; j_1, j_2\rangle$.

$$|j_1 - j_2| \leq j \leq j_1 + j_2$$

Note that $\hat{J}^2 |j, m, j_1, j_2\rangle = \hbar^2 j(j+1) |j, m, j_1, j_2\rangle$

$$\hat{J}^z |j, m, j_1, j_2\rangle = \hbar m |j, m, j_1, j_2\rangle$$

$$-j \leq m \leq j$$

Note that $[\vec{J}_1^2, \vec{J}^2] = [\vec{J}_1^2, (\vec{J}_1 + \vec{J}_2)^2] = 0$

(because for any $\vec{L}$: $[\vec{L}^2, L^{x,y,z}] = 0$
angular momentum squared commutes with every component of $\vec{L}$.)

$\Rightarrow$ these states are also eigenstates of $\hat{J}_1^2$ and $\hat{J}_2^2$: $\hat{J}_1^2 |j, m, j_1, j_2\rangle$
$= \hbar^2 j_1(j_1+1) |j, m, j_1, j_2\rangle$

Quite generally:

$$|j, m, j_1, j_2\rangle = \sum_{m = m_1 + m_2} C^{j_1 j_2 j}_{m_1 m_2 m} |j_1, m_1; j_2, m_2\rangle$$

$$\Downarrow$$

$\langle j_1, m_1; j_2, m_2 | j, m, j_1, j_2\rangle$ = Clebsch-Gordan coefficients

Explicit construction for $J_1 = J_2 = \frac{1}{2}$; $J_1 = 1$, $J_2 = \frac{1}{2}$; $J_1 = \ell$, $J_2 = \frac{1}{2}$.

① $J_1 = \frac{1}{2}$, $J_2 = \frac{1}{2}$ $\Rightarrow$ $0 = \frac{1}{2} - \frac{1}{2} \leq J \leq \frac{1}{2} + \frac{1}{2} = 1$

$\Rightarrow$ 4 states: 3 with $J = 1$ $(m = -1, 0, 1)$ and 1 with $J = 0$ $(m = 0)$.

<u>Original</u> states are $|\frac{1}{2}, \pm\frac{1}{2}; \frac{1}{2}, \pm\frac{1}{2}\rangle \xrightarrow[\text{simplify notation}]{} |\psi_{\pm\frac{1}{2}, \pm\frac{1}{2}}\rangle$

<u>New states</u> are eigenstates of total $\vec{J}^2$ and J_z (as well as $\vec{J}_1^2, \vec{J}_2^2$)

Denote as $|\phi_{j,m}\rangle$ for brevity.

Clearly $|\phi_{1,1}\rangle = |\psi_{\frac{1}{2}, \frac{1}{2}}\rangle$ as there is only 1 state with $m = 1 = \frac{1}{2} + \frac{1}{2} = m_1 + m_2$.

To find $|\phi_{1,0}\rangle$ we act with $\hat{J}_-$ on $|\phi_{1,1}\rangle$:

By def $\hat{J}_- |\phi_{1,1}\rangle = \sqrt{j(j+1) - m(m-1)}\, |\phi_{1,\underbrace{1-1}_{0}}\rangle = \sqrt{2}\, |\phi_{1,0}\rangle$

But in the old basis

$$\hat{J}_- |\phi_{1,1}\rangle = (\hat{J}_{1-} + \hat{J}_{2-}) |\psi_{\frac{1}{2}, \frac{1}{2}}\rangle = |\psi_{-\frac{1}{2}, \frac{1}{2}}\rangle + |\psi_{\frac{1}{2}, -\frac{1}{2}}\rangle$$

$$\Rightarrow |\phi_{1,0}\rangle = \frac{1}{\sqrt{2}} \left(|\psi_{-\frac{1}{2}, \frac{1}{2}}\rangle + |\psi_{\frac{1}{2}, -\frac{1}{2}}\rangle \right).$$

Then $\hat{J}_- |\phi_{1,0}\rangle = \sqrt{2 - 0}\, |\phi_{1,-1}\rangle = \sqrt{2}\, |\phi_{1,-1}\rangle$

but $\hat{J}_- |\phi_{1,0}\rangle = (\hat{J}_{1-} + \hat{J}_{2-}) \frac{1}{\sqrt{2}} \left(|\psi_{\frac{1}{2}, -\frac{1}{2}}\rangle + |\psi_{-\frac{1}{2}, \frac{1}{2}}\rangle \right) =$

$$= \frac{1}{\sqrt{2}}\, 2 |\psi_{-\frac{1}{2}, -\frac{1}{2}}\rangle = \sqrt{2}\, |\psi_{-\frac{1}{2}, -\frac{1}{2}}\rangle$$

$$\Rightarrow |\phi_{1,-1}\rangle = |\psi_{-\frac{1}{2}, -\frac{1}{2}}\rangle.$$

The remaining state $|\phi_{0,0}\rangle$ must be orthogonal to $|\phi_{1,0}\rangle$

$$\Rightarrow |\phi_{0,0}\rangle = \frac{1}{\sqrt{2}} \left(|\psi_{\frac{1}{2}, -\frac{1}{2}}\rangle - |\psi_{-\frac{1}{2}, \frac{1}{2}}\rangle \right).$$ <u>singlet</u>

by inspection

Naturally $|\phi_{1,m}\rangle$ = triplet states.

(2) Spin $1/2$ + Orbital angular momentum ℓ

$\vec{J} = \vec{L} + \vec{S}$. $\vec{S}^2 = \frac{3}{4}\hbar^2$

Old states are $|\ell, m_\ell ; s, m_s\rangle \Rightarrow |\psi_{m_\ell m_s}\rangle$

New states $|j, m ; \ell, s\rangle \Rightarrow |\phi_{jm}\rangle$, $m = m_\ell + m_s$

Two possibilities: $\begin{cases} j = \ell + \frac{1}{2} & \text{ladder contains } 2j+1 = \underline{2\ell+2} \text{ states} \\ j = \ell - \frac{1}{2} & \text{ladder contains } 2j+1 = 2\ell-1+1 = \underline{2\ell} \end{cases}$

$\Rightarrow$ total # states $2\ell+2+2\ell = 2(2\ell+1)$ which is exactly the dimension of $|\psi\rangle$ space.

Start with $\underline{\underline{j = \ell + \frac{1}{2}}}$: $|\phi_{\ell+\frac{1}{2}, m}\rangle = \sum_{m_\ell + m_s = m} c_{m_\ell m_s} |\psi_{m_\ell, m_s}\rangle =$

$$= \alpha_m |\psi_{m-\frac{1}{2}, \frac{1}{2}}\rangle + \beta_m |\psi_{m+\frac{1}{2}, -\frac{1}{2}}\rangle$$

$m_\ell = m - \frac{1}{2}$, $m_s = \frac{1}{2}$ $\qquad$ $m_\ell = m + \frac{1}{2}$, $m_s = -\frac{1}{2}$

Normalization: $|\alpha_m|^2 + |\beta_m|^2 = 1$

Apply lowering operator $\hat{J}_- |\phi_{\ell+\frac{1}{2}, m}\rangle \overset{\text{by def}}{=} \hbar\sqrt{j(j+1) - m(m-1)}\, |\phi_{\ell+\frac{1}{2}, m-1}\rangle$

(here $j = \ell + \frac{1}{2}$)

$$= \hbar\sqrt{(j+m)(j-m+1)}\, |\phi_{\ell+\frac{1}{2}, m-1}\rangle = \hbar\sqrt{(\ell+m+\tfrac{1}{2})(\ell-m+\tfrac{3}{2})} \cdot$$

$(j = \ell + \frac{1}{2})$

$$\cdot \left\{ \alpha_{m-1} |\psi_{m-\frac{3}{2}, \frac{1}{2}}\rangle + \beta_{m-1} |\psi_{m-\frac{1}{2}, -\frac{1}{2}}\rangle \right\} . \quad (1)$$

On the other hand

$$\hat{J}_- |\phi_{\ell+\frac{1}{2}, m}\rangle = (\hat{L}_- + \hat{S}_-) \left\{ \alpha_m |\psi_{m-\frac{1}{2}, \frac{1}{2}}\rangle + \beta_m |\psi_{m+\frac{1}{2}, -\frac{1}{2}}\rangle \right\}$$

$$= \hbar\, \alpha_m \sqrt{\ell(\ell+1) - (m-\tfrac{1}{2})(m-\tfrac{3}{2})} \left(|\psi_{m-\frac{3}{2}, \frac{1}{2}}\rangle + \sqrt{\tfrac{1}{2}(1+\tfrac{1}{2}) - \tfrac{1}{2}(\tfrac{1}{2}-1)}\, |\psi_{m-\frac{1}{2}, -\frac{1}{2}}\rangle \right)$$

$$+ \hbar\, \beta_m \sqrt{\ell(\ell+1) - (m+\tfrac{1}{2})(m-\tfrac{1}{2})} \left(|\psi_{m-\frac{1}{2}, -\frac{1}{2}}\rangle + \underbrace{\sqrt{\tfrac{1}{2}(\tfrac{1}{2}+1) - (-\tfrac{1}{2})(-\tfrac{1}{2}-1)}}_{\emptyset}\, \underbrace{|\psi_{m+\frac{1}{2}, -\frac{1}{2}-1}\rangle}_{\text{does not exist}} \right)$$

Use $l(l+1) - m(m-1) = (l+m)(l-m+1)$.

Thus $\hat{J}_- |\phi_{l+\frac{1}{2},m}\rangle = \hbar\alpha_m \sqrt{(l+m-\frac{1}{2})(l-m+\frac{3}{2})}\, |\psi_{m-\frac{3}{2},\frac{1}{2}}\rangle +$

$+ |\psi_{m-\frac{1}{2},-\frac{1}{2}}\rangle\Big) + \hbar\, \beta_m \sqrt{(l+m+\frac{1}{2})(l-m+\frac{1}{2})}\, |\psi_{m-\frac{1}{2},-\frac{1}{2}}\rangle$. (2)

Right-hand sides of (1) and (2) are the same $\Rightarrow$ we can compare coefficients:

$$\alpha_{m-1}\sqrt{(l+m+\tfrac{1}{2})(l-m+\tfrac{3}{2})} = \alpha_m \sqrt{(l+m-\tfrac{1}{2})(l-m+\tfrac{3}{2})} \qquad (3)$$

(The relation for β_m obtains from normalization: $|\beta_m|^2 = 1 - |\alpha_m|^2$.)

Observe that (3):

$$\frac{\alpha_m}{\sqrt{l+m+\frac{1}{2}}} = \frac{\alpha_{m-1}}{\sqrt{l+m-\frac{1}{2}}} = r \quad (r \text{ is constant independent of } m!)$$

Now, for a maximum possible $m = l+\frac{1}{2}$:

$$|\phi_{l+\frac{1}{2},l+\frac{1}{2}}\rangle = \alpha_{l+\frac{1}{2}} |\psi_{l,\frac{1}{2}}\rangle + \beta_{l+\frac{1}{2}} |\psi_{l+1,-\frac{1}{2}}\rangle$$

state with $m_l = l+1$, $m_s = -\frac{1}{2}$.

But $m_l \le l$ always

$\Rightarrow \boxed{\beta_{l+\frac{1}{2}} = 0}$

$$\Rightarrow \alpha_{l+\frac{1}{2}} = 1.$$

$$\Rightarrow r = \frac{\alpha_{l+\frac{1}{2}}}{\sqrt{l+l+\frac{1}{2}+\frac{1}{2}}} = \frac{1}{\sqrt{2l+1}}$$

$$\Rightarrow \alpha_m = \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}$$

$$|\beta_m|^2 = 1 - \frac{l+m+\frac{1}{2}}{2l+1} = \frac{l-m+\frac{1}{2}}{2l+1} \Rightarrow \beta_m = \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}$$

Thus $|\phi_{l+\frac{1}{2},m}\rangle = \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}\,|\psi_{m-\frac{1}{2},\frac{1}{2}}\rangle + \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}\,|\psi_{m+\frac{1}{2},-\frac{1}{2}}\rangle$

The states from the other ladder, with $\underline{\underline{j = l-\frac{1}{2}}}$;

must be orthogonal to $|\phi_{l+\frac{1}{2},m}\rangle$ at the same $\underline{m}$:

Thus

$$|\phi_{l-\frac{1}{2},m}\rangle = \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}\,|\psi_{m-\frac{1}{2},\frac{1}{2}}\rangle - \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}\,|\psi_{m+\frac{1}{2},-\frac{1}{2}}\rangle$$

Clearly $\langle \phi_{l+\frac{1}{2},m} | \phi_{l-\frac{1}{2},m}\rangle = 0$ always.

Thus we found Clebsch-Gordan coefficients for $\vec{J} = \vec{L} + \vec{S}$, $S = \frac{1}{2}$.

They are $\alpha_m = \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}$ and $\beta_m = \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}$.